

ME315A_LAB07

Letícia Santos Tavares

CONEXÃO R »> SQLite

1. Baixe o arquivo `uwmadison.sqlite3`. Conecte-se a ele usando o pacote `RSQLite`, armazenando a conexão em uma variável `conn`.

```
library(RSQLite)
library(DBI)
conn <- dbConnect(SQLite(), 'uwmadison.sqlite3')
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (`subjects.abbreviation='STAT'`);

QUEM SÃO

```
SELECT DISTINCT instructors.name AS STAT_TEACHERS
FROM subjects
LEFT JOIN subject_memberships
ON subjects.code= subject_memberships.subject_code
LEFT JOIN sections
ON subject_memberships.course_offering_uuid = sections.course_offering_uuid
LEFT JOIN teachings
ON sections.uuid = teachings.section_uuid
LEFT JOIN instructors
ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
```

Table 1: Displaying records 1 - 10

STAT_TEACHERS
JINGCI MENG
RICHARD A. JOHNSON

STAT_TEACHERS
KARL ROHE
ZHENGJUN ZHANG
ERIK NORDHEIM
PAUL RATHOUZ
RICHARD J. CHAPPELL
BRET HANLON
BRET LARGET
CECILE ANE

QUANTOS SÃO:

```
SELECT COUNT(DISTINCT instructors.name) AS STAT_TEACHERS_TOTAL
FROM subjects
LEFT JOIN subject_memberships
ON subjects.code= subject_memberships.subject_code
LEFT JOIN sections
ON subject_memberships.course_offering_uuid = sections.course_offering_uuid
LEFT JOIN teachings
ON sections.uuid = teachings.section_uuid
LEFT JOIN instructors
ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
```

Table 2: 1 records

STAT_TEACHERS_TOTAL
377

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):
 - Quem é o professor mais difícil?

```
SELECT DISTINCT instructors.name, ((4.0*a_count + 3.5*ab_count + 3.0*b_count + 2.5*bc_count +
FROM grade_distributions
LEFT JOIN sections
ON grade_distributions.course_offering_uuid = sections.course_offering_uuid
```

```

LEFT JOIN teachings
ON sections.uuid = teachings.section_uuid
LEFT JOIN instructors
ON teachings.instructor_id = instructors.id
LEFT JOIN subject_memberships
ON subject_memberships.course_offering_uuid = grade_distributions.course_offering_uuid
LEFT JOIN subjects
ON subjects.code = subject_memberships.subject_code
WHERE subjects.abbreviation='STAT'AND (a_count + ab_count + b_count + bc_count + c_count + d_count) > 0

ORDER BY GPA
LIMIT 1

```

Table 3: 1 records

name	GPA
CHENXI LI	2.30531

- Quem é o professor mais fácil?

```

SELECT DISTINCT instructors.name, ((4.0*a_count + 3.5*ab_count + 3.0*b_count + 2.5*bc_count + 2.0*cd_count + 1.0*de_count) / (a_count + ab_count + b_count + bc_count + c_count + d_count)) AS GPA
FROM grade_distributions
LEFT JOIN sections
ON grade_distributions.course_offering_uuid = sections.course_offering_uuid
LEFT JOIN teachings
ON sections.uuid = teachings.section_uuid
LEFT JOIN instructors
ON teachings.instructor_id = instructors.id
LEFT JOIN subject_memberships
ON subject_memberships.course_offering_uuid = grade_distributions.course_offering_uuid
LEFT JOIN subjects
ON subjects.code = subject_memberships.subject_code
WHERE subjects.abbreviation='STAT'AND (a_count + ab_count + b_count + bc_count + c_count + d_count) > 0

ORDER BY GPA DESC
LIMIT 1

```

Table 4: 1 records

name	GPA
DUZHE WANG	4

- Qual é a disciplina mais difícil?

```
SELECT DISTINCT subjects.name, ((4.0*a_count + 3.5*ab_count + 3.0*b_count + 2.5*bc_count + 2
FROM grade_distributions
LEFT JOIN sections
ON grade_distributions.course_offering_uuid = sections.course_offering_uuid
LEFT JOIN teachings
ON sections.uuid = teachings.section_uuid
LEFT JOIN instructors
ON teachings.instructor_id = instructors.id
LEFT JOIN subject_memberships
ON subject_memberships.course_offering_uuid = grade_distributions.course_offering_uuid
LEFT JOIN subjects
ON subjects.code = subject_memberships.subject_code
WHERE (a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) > 0

ORDER BY GPA
LIMIT 1
```

Table 5: 1 records

name	GPA
PHYSICAL EDUC ACTIVITY PROGM	0

- Qual é a disciplina mais fácil?

```
SELECT DISTINCT subjects.name, ((4.0*a_count + 3.5*ab_count + 3.0*b_count + 2.5*bc_count + 2
FROM grade_distributions
LEFT JOIN sections
ON grade_distributions.course_offering_uuid = sections.course_offering_uuid
LEFT JOIN teachings
ON sections.uuid = teachings.section_uuid
LEFT JOIN instructors
ON teachings.instructor_id = instructors.id
LEFT JOIN subject_memberships
ON subject_memberships.course_offering_uuid = grade_distributions.course_offering_uuid
```

```

LEFT JOIN subjects
ON subjects.code = subject_memberships.subject_code
WHERE (a_count + ab_count + b_count + bc_count + c_count + d_count + f_count) > 0

ORDER BY GPA DESC
LIMIT 1

```

Table 6: 1 records

name	GPA
Biomedical Engineering	4

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-22 17:31:19 -03"
```